Part E · Justification for the clinical review figure

Data: vitals_10k.csv (10,000 admissions, 1 Jan–30 Apr 2026), synthetic, generated for CDB601220.

1 · Each panel: question, chart type, Week 2 channel rule

Panel 1, is systolic pressure one population? A histogram, because one continuous variable is read by bar length from a shared zero baseline, the most accurate channel (position/length on a common scale). It shows two humps, near 120 and 160 mmHg (58% / 42% split at the 140 valley); nothing else in the file (ward, sex, age) separates the groups, so I report it as unexplained.

Panel 2, how do wards differ in length of stay? A box plot, because it compares one distribution across four categories with median, spread and tail in one mark, ordered by median so position carries the ranking. Log scale because the data are right-skewed. ICU median 3.47 days against 0.25 for day surgery (about 14x). One colour with a single highlighted category (ICU).

Panel 3, how does cholesterol move with age? A scatter, because two continuous variables need position on both axes. It is the one pair worth plotting: panel 4 shows age–cholesterol ($\rho = 0.33$) as the only association that is neither arithmetic nor flat. Alpha 0.18 handles 10,000 overlapping points without hiding density; the line is OLS, slope +0.21 mmol/L per decade.

Panel 4, which variables move together? A heatmap, as a screening tool across 36 pairs. Colour is the weakest channel, so every cell is annotated; a diverging RdBu_r scale pinned to −1…+1 is used because a correlation has a sign, with the redundant upper triangle masked. Palette: the three-slot Okabe–Ito set (worst CVD ΔE 8.4, tritan); the 4th slot is not used.

2 · Every parameter chosen, and why

Panel 1: bin width 4 mmHg (whole mmHg, so bins never alias; Freedman–Diaconis gives 3.8; any width from 2 to 20 shows the same two peaks); split at 140 mmHg, the valley. **Panel 2:** log y-axis (stays are skewed; 2 zero-day stays cannot be drawn on it and are left off that axis only); whiskers at the default 1.5 IQR; fliers shown (*showfliers* is left on). **Panel 3:** alpha 0.18, marker size 5, no subsampling. **Panel 4:** Spearman, not Pearson, because glucose and stay length are skewed and, as case 6 showed, one extreme value can fake a Pearson r; vmin −1, vmax +1, centre 0, annotation to two decimals; *map_mmhg* excluded because it is computed from systolic and diastolic. Export: 300 dpi, tight bbox, white facecolor.

3 · Every row removed, with rule, count and effect

Rule	Count	Mean moves
Ward: 12 spellings conformed to 4 (no rows lost)	0 dropped	n/a
Dates: 3 formats parsed, day-first for slash dates (first field reaches 31)	0 dropped	n/a
BMI sentinels 0 and 999 set to missing (row kept)	486 (84 + 402)	66.18 → 27.35 kg/m²
Heart rate < 30 or > 250 bpm set to missing: device faults (row kept)	37 (8 zeros, 29 of 331–415)	78.83 → 78.05 bpm
Length of stay > 100 days set to missing (next-highest value is 24)	18 (140–274 days)	2.31 → 1.95 days

Ethics clause. Readings above 250 bpm, and readings of 0 bpm, were removed as device faults (n = 37). Mean heart rate moves from 78.83 to 78.05 bpm. The figure was also produced without the exclusions (heart rate, BMI and stay length); it is in the appendix (partD_figure_appendix.png) and its four messages are the same, because panel 4 uses rank correlations. Fault readings are blanked rather than deleted, so no patient is lost.

Symmetry test. The thresholds come from physiology and from visible gaps in the data (stays jump from 24 to 140 days; heart rates run 38–135 bpm, then jump to 331+, with a block of zeros), not from the figure. I would have applied the same rules if they had weakened my conclusion; here they happen not to change it. **Not done:** *risk_score* is a ratio (events/day) and is not averaged by ward, because a mean of ratios is not a rate.